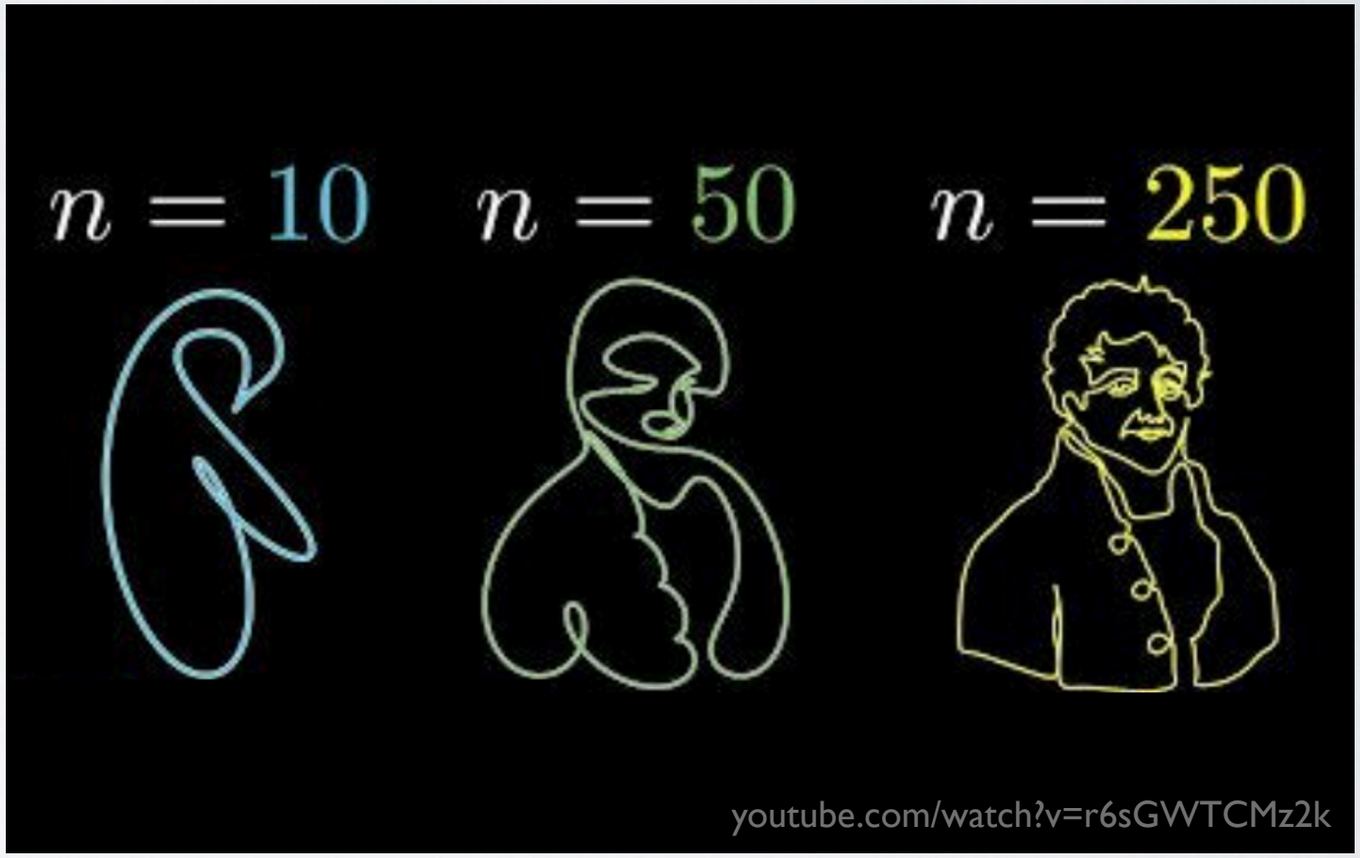
Week 9 - #2

Fourier Series and Transforms (II)





Today: Ch 7

Next Class: Ch 8

Ji-hoon Kim (Seoul National University)

This material is for the educational purpose only, and is not for public consumption or mass distribution without the author's explicit consent.
Despite his best efforts, the author apologizes for any inaccurate or outdated information and references.

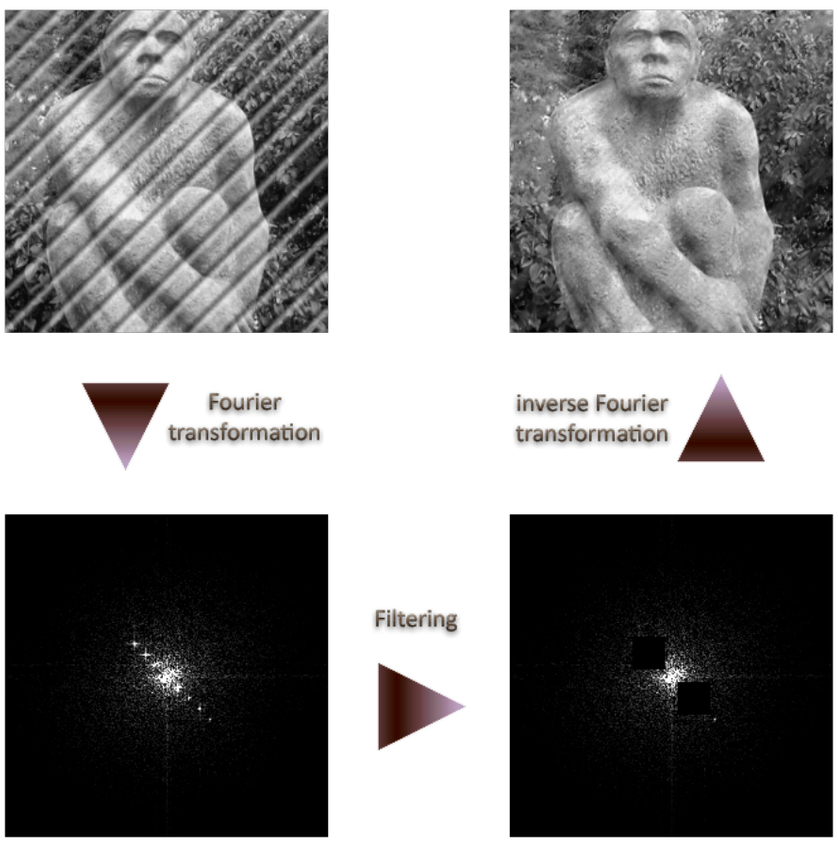
Rudimentary Mathematical Methods of Physics (Fall 2026): Quiz #13

— [open book and open note, "and" cellphone or laptop, drop it off as you leave the class] —

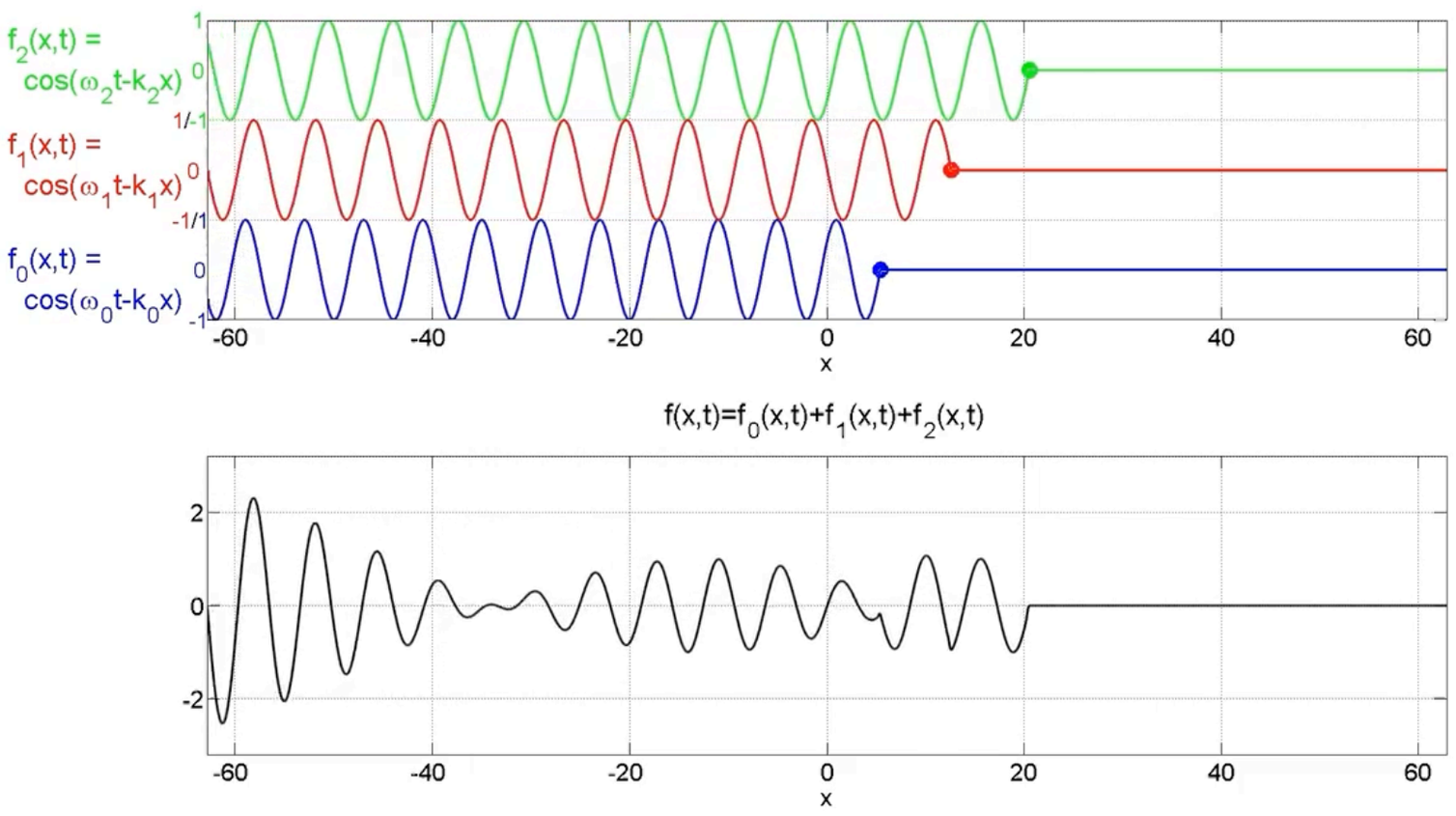
Please write down your name and student ID in the top right corner. (0.0 pt: no paper found with your name / 0.2 pt: paper found with your name and some answers / 0.5 pt: good answers)

1. Boas, Chapter 7, Section 5, Problem 2

2. Briefly explain, in layman's terms and without using detailed math, what each of the figure/movie below is trying to visualize.



Fourier filtering, Wikipedia commons



youtube.com/watch?v=ePJdV75fT5o

Rudimentary Mathematical Methods of Physics (Fall 2026): Suggested Problems in Chapter 8, Boas, 3rd ed.

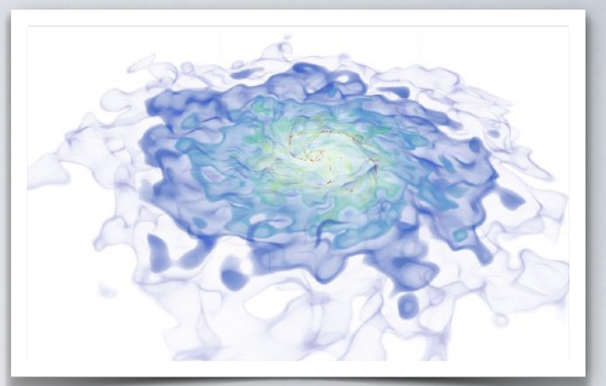
The problems I suggest you to take a deeper look into include, but are not limited to, the following. The class homework assignments will mainly be from this list.

- Section 01: Problems 4
- Section 02: Problems 6, 8, 11, 12, 20(b), 34
- Section 03: Problems 4, 11, 14, 16, 20
- Section 04: Problems 2, 6, 8, 10, 15, 20
- Section 05: Problems 6, 9, 11, 13, 15, 24, 31(c)(d)
- Section 06: Problems 5, 8, 10, 14, 21, 23, 28, 34, 35, 42
- Section 07: Problems 2, 3, 8, 17, 22, 30
- Section 08: Problems 1, 2, 4, 7, 10, 11, 12, 19, 22, 26
- Section 09: Problems 8, 9, 14, 16, 28, 30, 34, 38
- Section 10: Problems 3, 8, 11, 14, 18
- Section 11: Problems 6(b)(c), 7, 9, 15, 21, 23, 25
- Section 12: Problems 2, 4, 6, 11, 16

<u>Chapter 13, Sections 1-4</u>

- Section 01: Problems 1, 4
- Section 02: Problems 1, 2, 3, 7
- Section 03: Problems 2, 3, 7
- Section 04: Problems 2, 5

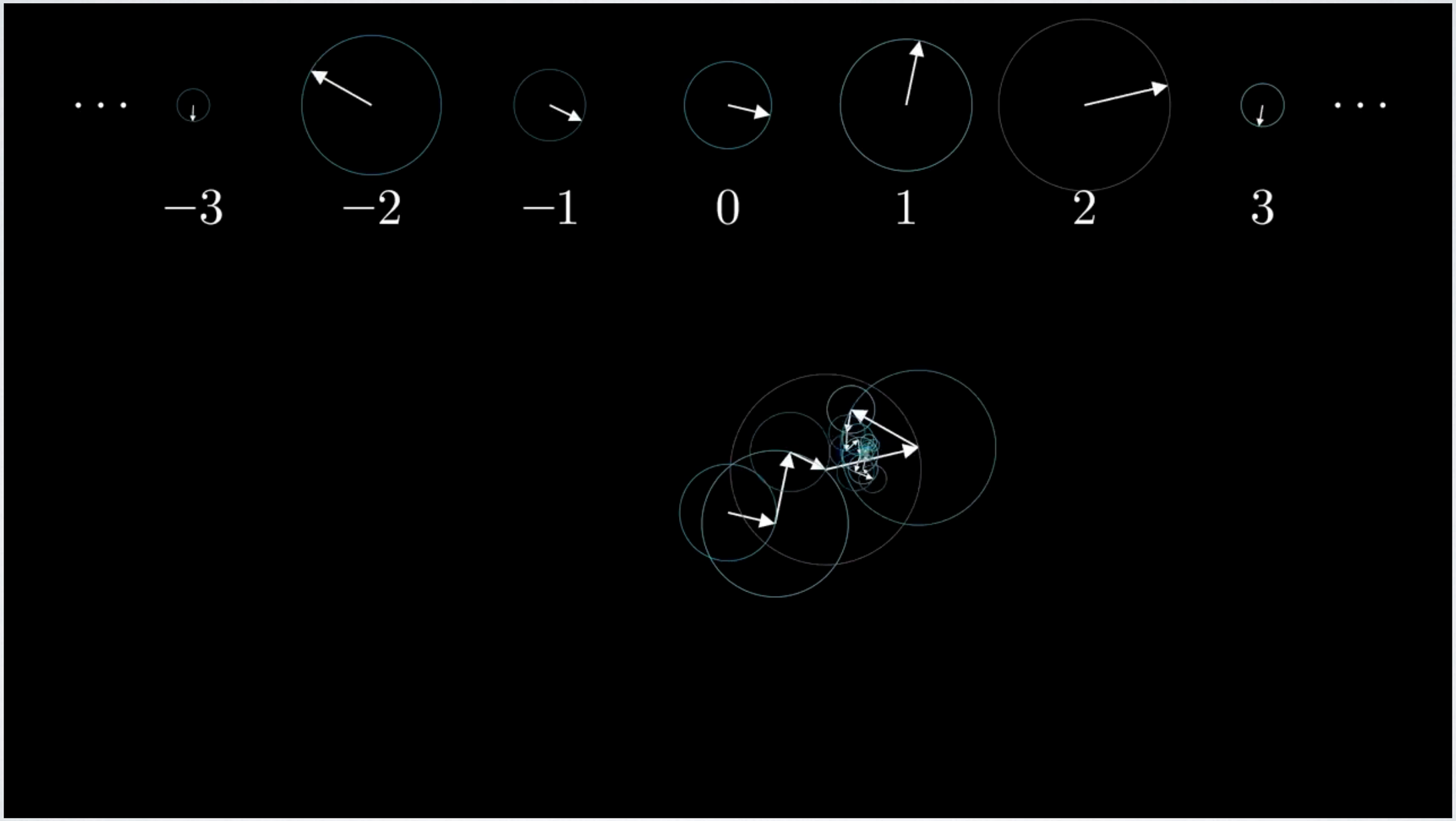
76 problems / 383 total



Complex Fourier Series

Complex Fourier Series

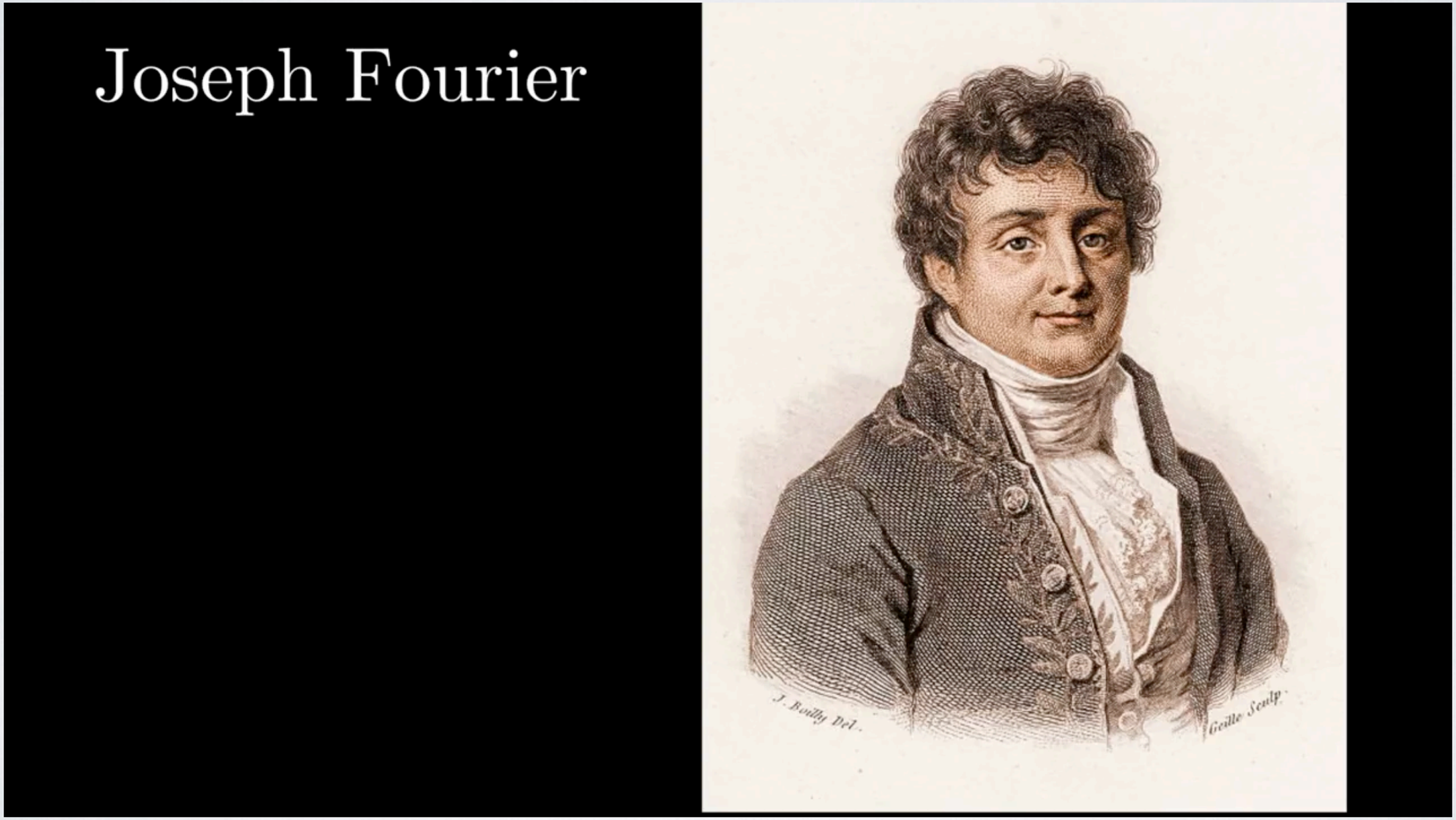
- A visualization of complex Fourier series.



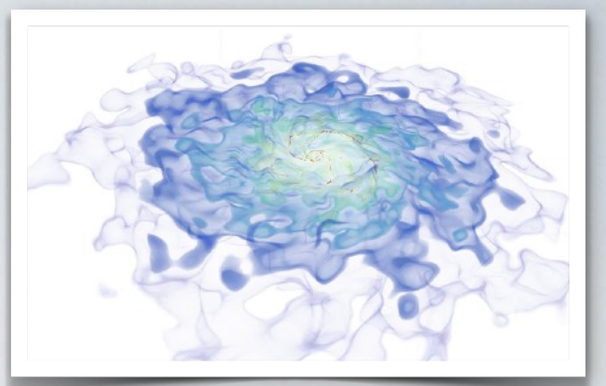
youtube.com/watch?v=r6sGWTCMz2k

Complex Fourier Series

- A visualization of complex Fourier series.



youtube.com/watch?v=r6sGWTCMz2k

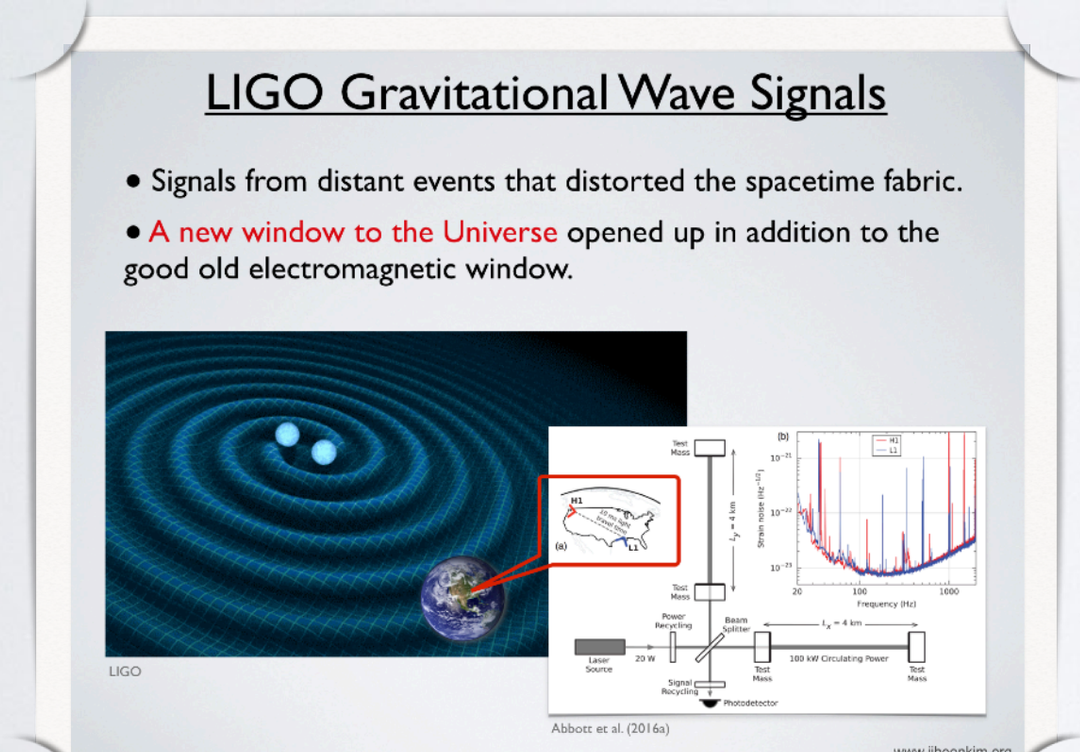


Term Project Paper

Term Project: Topic & Progress Check!

- Post your topic (+ answer to "What makes your project original?") and 2-page "progress report" slides as images by 23:00 pm, Oct. 27 (Tue) on a shared Google doc:

https://docs.google.com/presentation/d/1t2kbdpGYGkcmJx5QXF4JSP7JY1xjv1kawIrbmpU_2k0



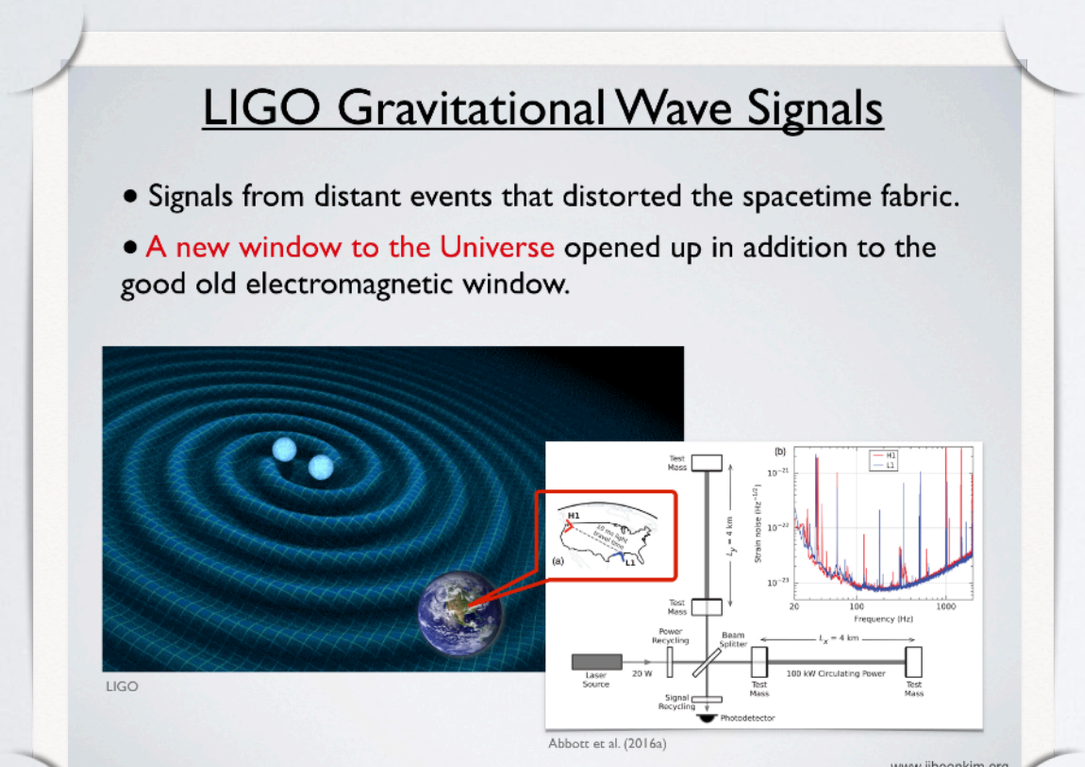
Extra credit guideline

- Additional +2: awarded to 2-3 students who were voted the best by your peers, and to 2-3 students who provided the most useful feedback.

Vote For The Best By Tomorrow (Oct. 29)

- Please look through the submitted slides, and vote for the most interesting, promising, and original progress report!

https://docs.google.com/presentation/d/1t2kbdpGYGkcmJx5QXF4JSP7JY1xjv1kawIrbmpU_2k0



Extra credit guideline

- Additional +2: awarded to 2-3 students who were voted the best by your peers, and to 2-3 students who provided the most useful feedback.

Term Project: Topic & Progress Check!

- A total of 66 students submitted the topic & progress slides!

- [ ] 장은율 / Fractal Network: 배터리 전극 내의 다공
- [ ] 김민찬 / Which stochastic physical process ex
- [ ] 염동현 / The mathematical structure of an ide
- [ ] 오승민 / Interactive star chart + quiz program
- [ ] 최재민 / Is the Spin Gravity of Project Hail Mar
- [ ] 천정환 / Why Does a Vibrating Smartphone Cr
- [ ] 오경서 / LHCII exciton model의 단순화에 따른 흡
- [ ] 정수호 / 허수의 이용에 대한 실질적인 물리량으로의
- [ ] 손건희 / 테일러 급수와 다항식 근사를 이용한 Retar
- [ ] 손정우 / 관성 텐서를 이용한 테슬라 로드스터의 분석
- [ ] 신건 / <길>(La Strada)에 등장하는 쇠사슬 차력의
- [ ] 박정준 / Transformations of Electric and Magn
- [ ] 정승민 / 가로등의 거울 곡면 형상 설계
- [ ] 윤지형 / Can Mathematics Tell the Difference f
- [ ] 정상원 / 만화 '주술 회전' 속 무하한 배리어의 물리학
- [ ] 이동하 / Nanopore Ion Transport에서 입자 내부
- [ ] 손명빈 / 국내 증권사의 S&P500 etf 상품의 기대 수
- [ ] 김가은 / SF 소설 < 모든 사람에 대한 이론(이하진)
- [ ] 정종인 / 황제펭귄 무리의 이동 파동에 대한 물리적 분
- [ ] 김도훈 / Elevator Traffic Optimization Using Sim
- [ ] 문성진 / The process of same word becomes di
- [ ] 안준혁 / Wavelet concentration as an energy foc
- [ ] 양태양 / 기술적 지표는 시장을 얼마나 잘 읽는가? - 푸
- [ ] 김진형 / Application of "Fragility Score" in Ches

- [ ] 박재현 / Active Noise Cancellation, but in Spac
- [ ] 박동선 / 인간의 발성 및 청각 구조만으로 주변 사물
- [ ] 양현서 / 잡음은 약한 소리의 검출을 도울 수 있을까?
- [ ] 신현빈 / 하트 모양의 아인슈타인 고리를 만들 수 있
- [ ] 박준수 / 최소환기문제에 대한 답을 미분방정식으로
- [ ] 박승범 / 맨몸운동 Lever 동작의 역학적 분석
- [ ] 유정원 / Disperser+ : Designing Arbitrary Grou
- [ ] 신준서 / 특정 해상풍력단지의 비-주기적 풍속 변화
- [ ] 황서원 / 초기 금속량에 따른 DAV의 g-mode 주기
- [ ] 이동규 / Predicting a Hidden Fifth Force in Cha
- [ ] 나중호 / Model a 3-Dimensional Space Billiard
- [ ] 장지호 / 집단 유체화 현상과 관련한 거시적 경험 모
- [ ] 이민정 / 지하철에서의 '빠른 하차'의 모순: 정차시간
- [ ] 김도윤 / 여러 전자가 함께 진동하면, 분극이 서로 상
- [ ] 최민석 / Fourier-Based Analysis of Spectral an
- [ ] 김가연 / Is There an Express Fee for Saving Ea
- [ ] 박준석 / AI 시대에 물리학자로 살아남기: 공간적으
- [ ] 성찬영 / 유체 모델을 이용한 지하철 역사 내 승객 이
- [ ] 남정우 / 일렉기타 피킹 하모닉스에서 피킹 위치와 입
- [ ] 차명진 / Can a Local Misjudgment Trigger a Ga
- [ ] 명재호 / 해류, 바람모델 분석을 통한 오기기아의 위치
- [ ] 유현석 / 에드거 앨런 포의 「아몬틸라도의 술통」 속 지하
- [ ] 고예준 / 스파이더맨 웹 스윙의 역학적 분석
- [ ] 이윤성 / 2차원 또는 3차원 공간 상에서 병력 흐름과
- [ ] 민지호 / Finding Internal Kinematics and Rotat
- [ ] 구준모 / 바이러스의 치명율, 감염성, 회복기간, 그리
- [ ] 김성윤 / Effects of the Temporal Ordering of Vi
- [ ] 정상교 / Analysis of representative seed gener
- [ ] 김승범 / 스파이더맨이 떠나고 난 뒤의 줄의 움직임.
- [ ] 안진식 / analizing the sound of water boiling b
- [ ] 최재웅 / 자취방에서 베이스가 웅웅거리는 이유를 방
- [ ] 정찬혁 / 정중신경에 작용하는 평균 일률 분석 기반의
- [ ] 유진서 / 수분 조절 능력을 통한 냉각 에너지의 물리적
- [ ] 황지성 / Experimental and Computational Invest
- [ ] 이창현 / Why Can Chopsticks Pick Up Slippery F
- [ ] 안재영 / How can a pedestrian survive from car
- [ ] 이연준 / 백색왜성 쌍성계에서 공전에 의한 질량 유출
- [ ] 유종선 / 각종 이산적인 것들의 푸리에 변환
- [ ] 송은호 / How does covering a cup affect the co
- [ ] 최준 / 양자역학을 사용해 구현할 수 있는 마술 혹은 마
- [ ] 민형준 / 심적 회전 속도를 고려하여 무작위 회전 제어
- [ ] 권순현 / H_2 T^+ 의 β- 붕괴에 따른 축소차원 핵 양자

Vote For The Best By Tomorrow (Oct. 29)

- For the remaining 0.5 point for your attendance score today.
- Go to jihoonkim.org → Teaching → Progress Check "Vote"
- Or: https://forms.gle/hdEbzdpD3vvBQeVLA

Week 03-2. Linear Algebra

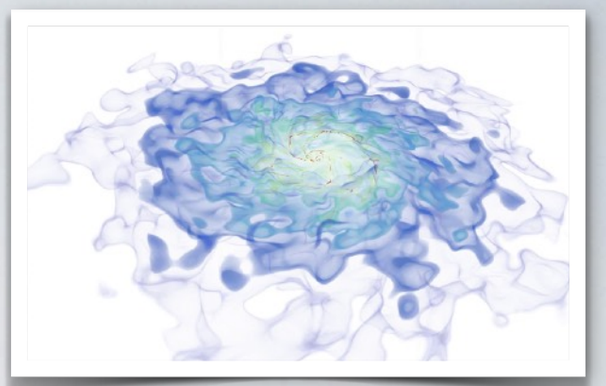
Week 04-1. Linear Algebra

Week 04-2. Linear Algebra + Term Project Paper How-To

Week 05-1. Linear Algebra, Tensor Analysis (I) + 2nd Homework (pdf) + Solution (pdf)

Week 05-2. Partial Differentiation + Term Project Progress Check (slides) + Vote (link)

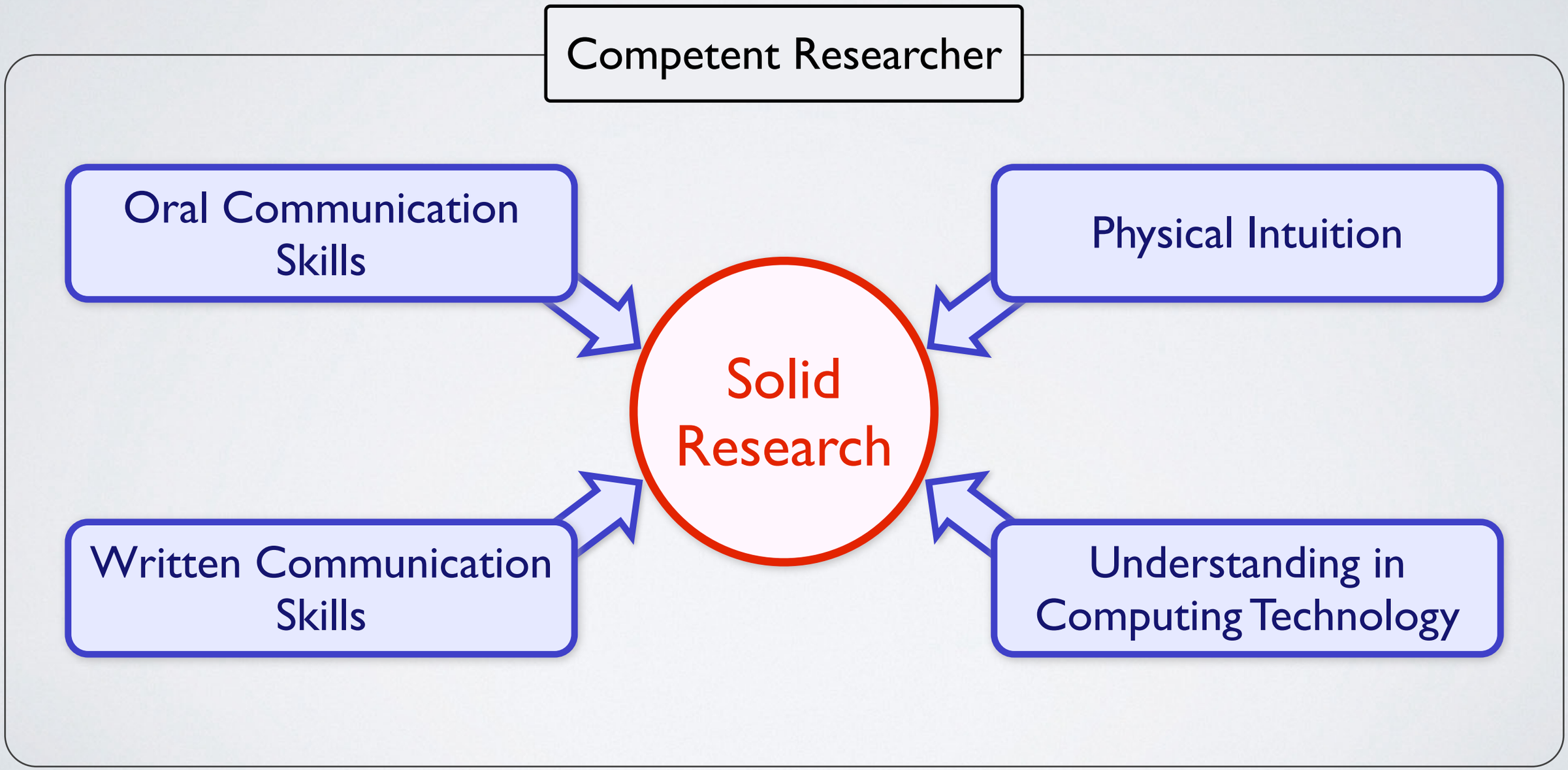
- Select 2 students you find promising - **not** yourself!



Term Project Presentation

Essential Skillsets For (Astro)Physicists

- To become proficient and effective researchers while relying less on guesswork as you navigate through the academia



Essential Skillsets For (Astro)Physicists

Competent Researcher

Oral Communication Skills

- How to present long and short talks
- How to answer questions
- How to make an elevator pitch

Written Communication Skills

- How to write papers and proceedings
- How to write referee reports
- How to respond to referee reports

Solid Research

Physical Intuition

Understanding in Computing Technology

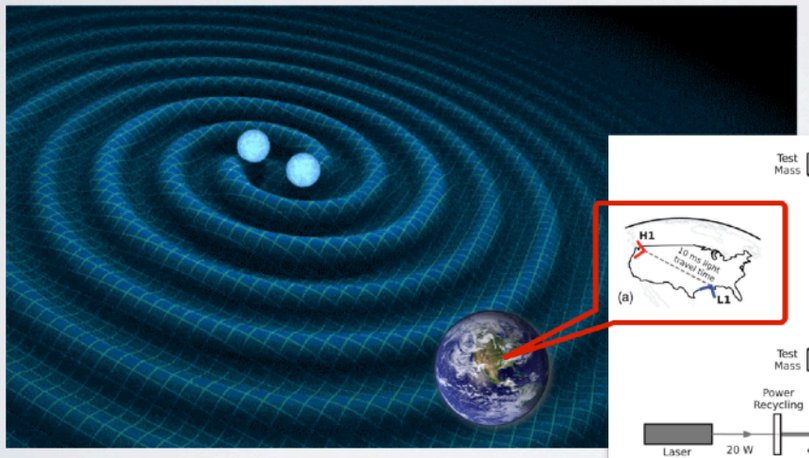


Term Project Presentation (Dec. 7)

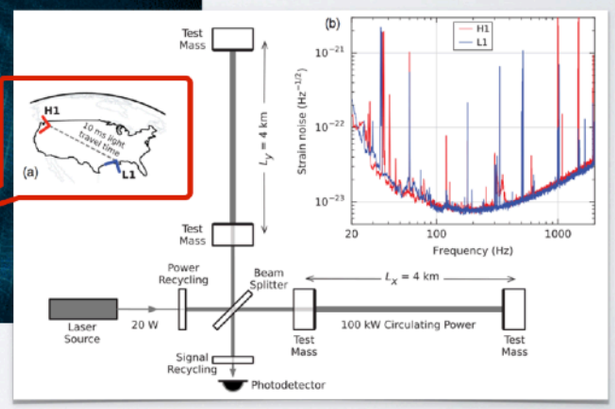
- 10-min presentation (+1-min Q&A) with maximum 6-page slides (including the title page) to present your term project.

LIGO Gravitational Wave Signals

- Signals from distant events that distorted the spacetime fabric.
- A new window to the Universe opened up in addition to the good old electromagnetic window.



LIGO



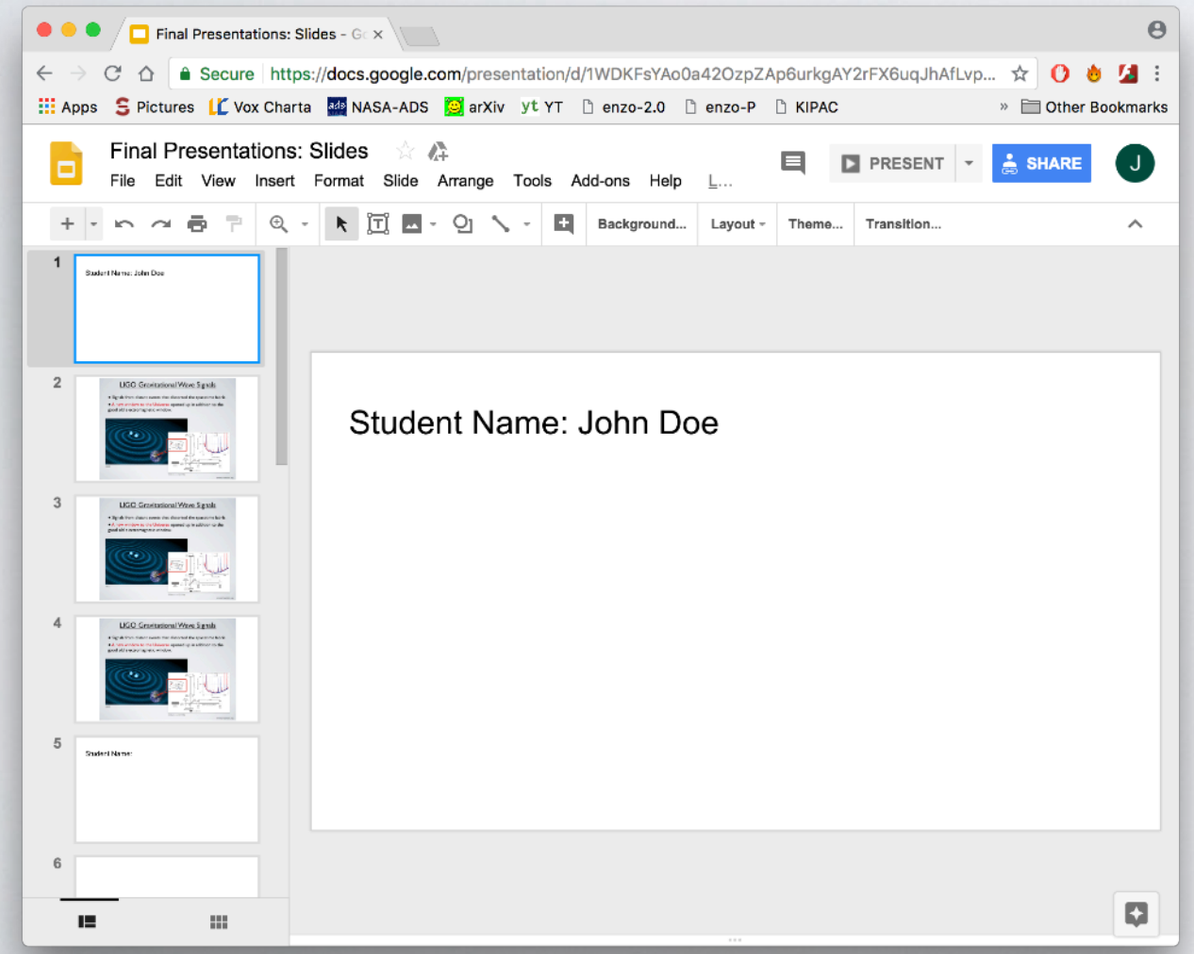
Abbott et al. (2016a)

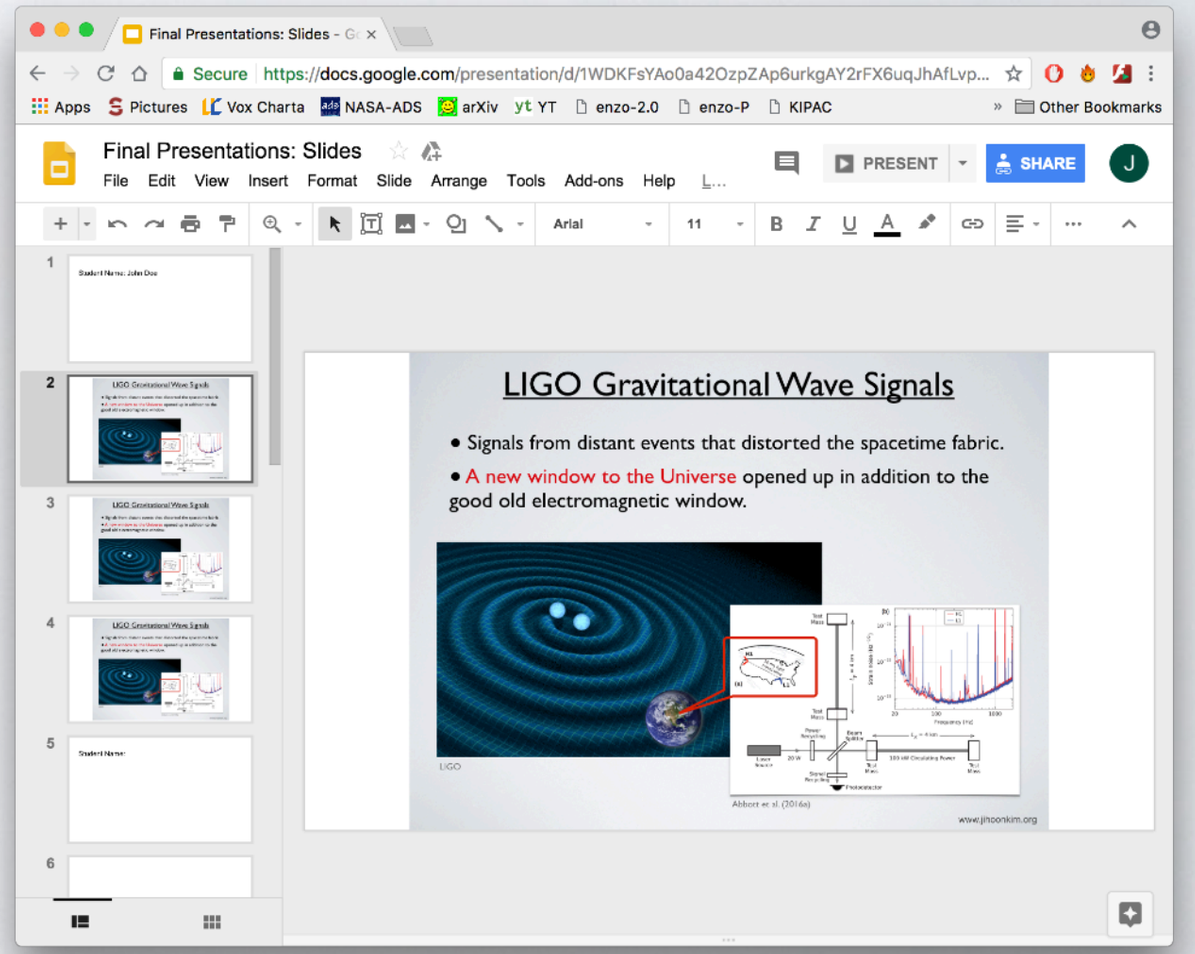
www.jihoonkim.org

Term Project Presentation (Dec. 7)

- Drag and drop your slides as images by 23:00pm, Nov. 28 (Sat) on a shared Google document below:

https://docs.google.com/presentation/d/1oPyPm5wrFonzy9C9VqLr--_uYQn2SVa4zQinMHnVnak

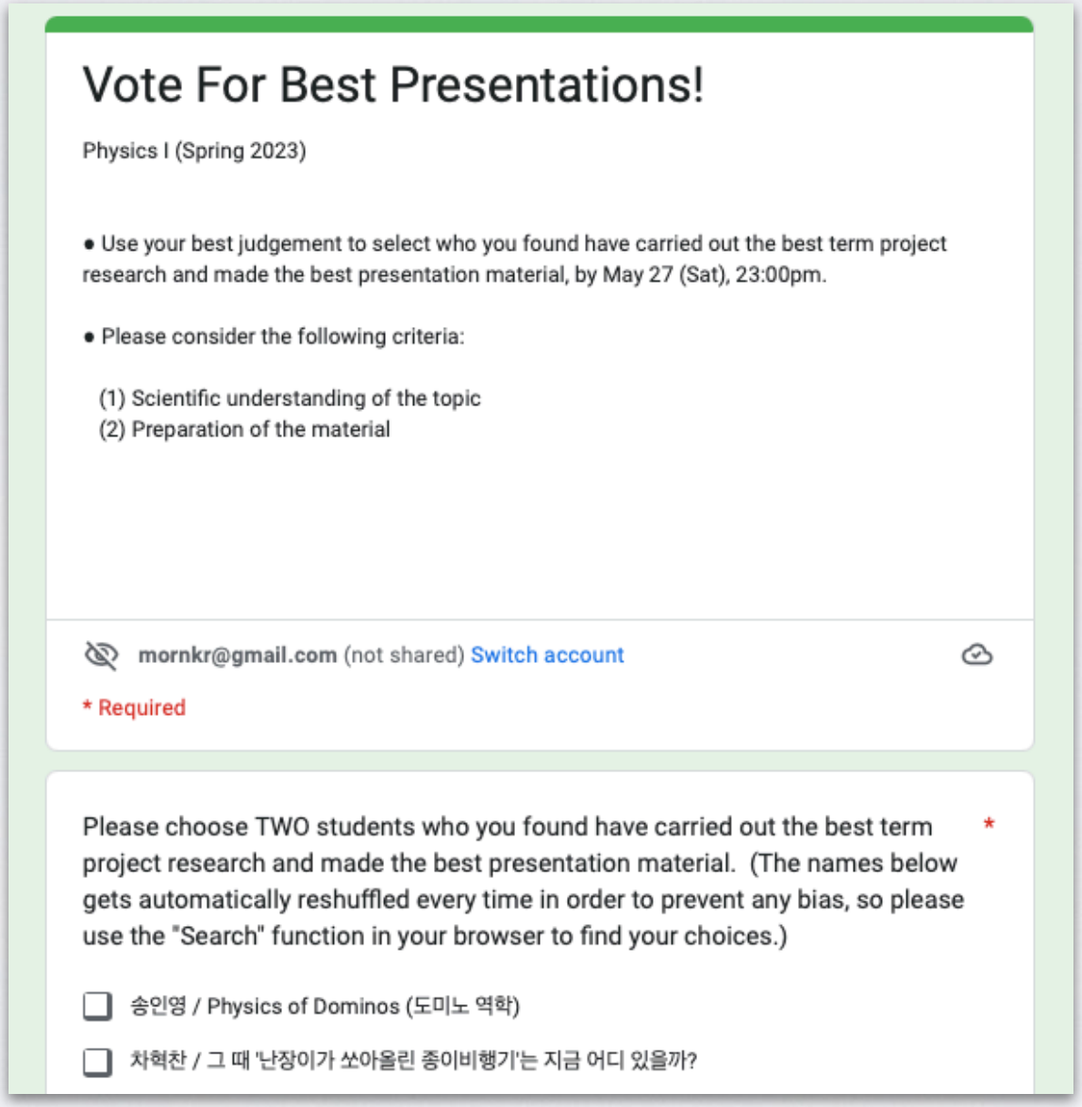




Term Project Presentation (Dec. 7)

- Please think of this not as an exam, but as a time to introduce your fellow student to an interesting topic you want to share.

- Your presentation material will be evaluated by your peers.



Vote For Best Presentations!

Physics I (Spring 2023)

• Use your best judgement to select who you found have carried out the best term project research and made the best presentation material, by May 27 (Sat), 23:00pm.

• Please consider the following criteria:

(1) Scientific understanding of the topic
(2) Preparation of the material

mornkr@gmail.com (not shared) Switch account

* Required

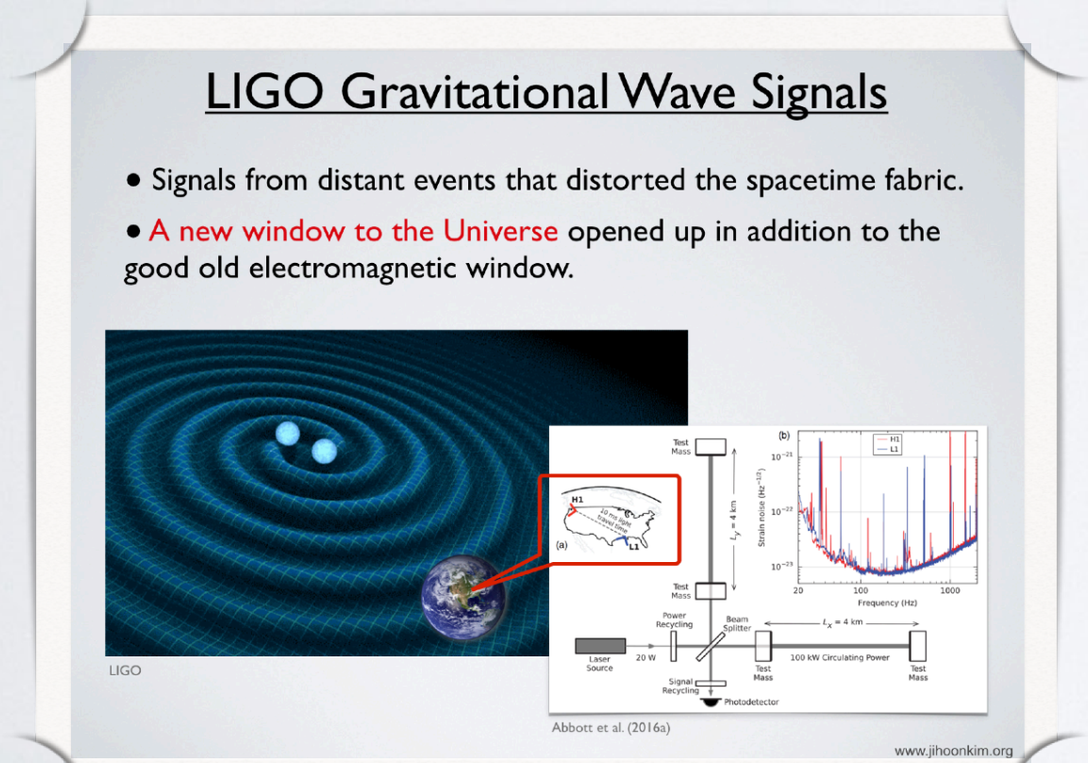
Please choose TWO students who you found have carried out the best term project research and made the best presentation material. (The names below gets automatically reshuffled every time in order to prevent any bias, so please use the "Search" function in your browser to find your choices.) *

송인영 / Physics of Dominos (도미노 역학)

차혁찬 / 그 때 '난장이가 쏘아올린 종이비행기'는 지금 어디 있을까?

Term Project Presentation (Dec. 7)

- 10-min presentation (+1-min Q&A) with maximum 6-page slides (including the title page) to present your term project.



Extra credit guideline

- +2: for volunteering for a presentation by submitting your slides before Nov. 28 (even if we have too many volunteers you will get +2 just for submitting the slides).

- Additional +2: awarded to 4-5 students who were voted the best by your peers, and to 2-3 students who provided the most useful feedback.

5-Minute Talk for Students in Conferences

Coffee break & poster session		15:20-15:40	
Observations of AGN Feedback (IV) Chair: Yingjie Peng	The most obscured galaxy nuclei – dusty AGNs and collimated molecular outflows	15:40-16:05	Susanne Aalto (remote)
	Observational Clues about Feedback in the Circumgalactic Medium of Massive Galaxies	16:05-16:30	Megan Donahue
	Mapping the Circumgalactic Medium Kinematics Through the Diffuse Emission	16:30-16:45	Zheng Cai
	Density diagnostics of ionized winds driven away from black holes	16:45-17:00	Junjie Mao (remote)
	The complex (negative and positive) AGN feedback from outflows and jets in the Teacup	17:00-17:15	Giacomo Venturi
	Outflows in AGN host galaxies from MaNGA	17:15-17:20	Kateryna Vovk
	The impact of AGN activity in nearby Seyfert galaxies: high resolution study of the multiphase ISM	17:20-17:25	Maria Vittoria Zanchettin
	Intermediate mass black hole feedback in dwarf galaxy simulations with a resolved interstellar medium and accurate nuclear stellar dynamics	17:25-17:30	Christian Partmann
	AGN feedback in the early Universe	17:30-17:55	Roberto Maiolino (remote)

3-Minute Thesis @ Caltech

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3MT: Three Minute Thesis

The Three Minute Thesis competition challenges Caltech graduate students to explain their research in an engaging and clear three-minute talk intended for a non-specialist audience. Given Caltech's prominence in science and engineering communities, we feel it's important to give Caltech graduate students the opportunity to practice communicating their science in ways that will resonate and inspire.

3MT® THREE MINUTE THESIS
FOUNDED BY THE UNIVERSITY OF QUEENSLAND





Competition Information

Prizes

There will be four prizes awarded:

- **$3,000** for first place
- **$2,000** for second place
- **$1,000** for third place
- **$500** for the audience favorite as chosen by attendees

Current Events
- On Display
- 3MT: Three Minute Thesis
- Stress Free Finals Week
- Virtual Library Week
- Senior Thesis Prize
- Open Access Week

Key Dates

- **March 1, 2023**: Submissions Open
- **March 22, 2023**: Submissions Due
- **April 10, 2023**: Final Selection
- **April 25, 2023**: 3MT Competition

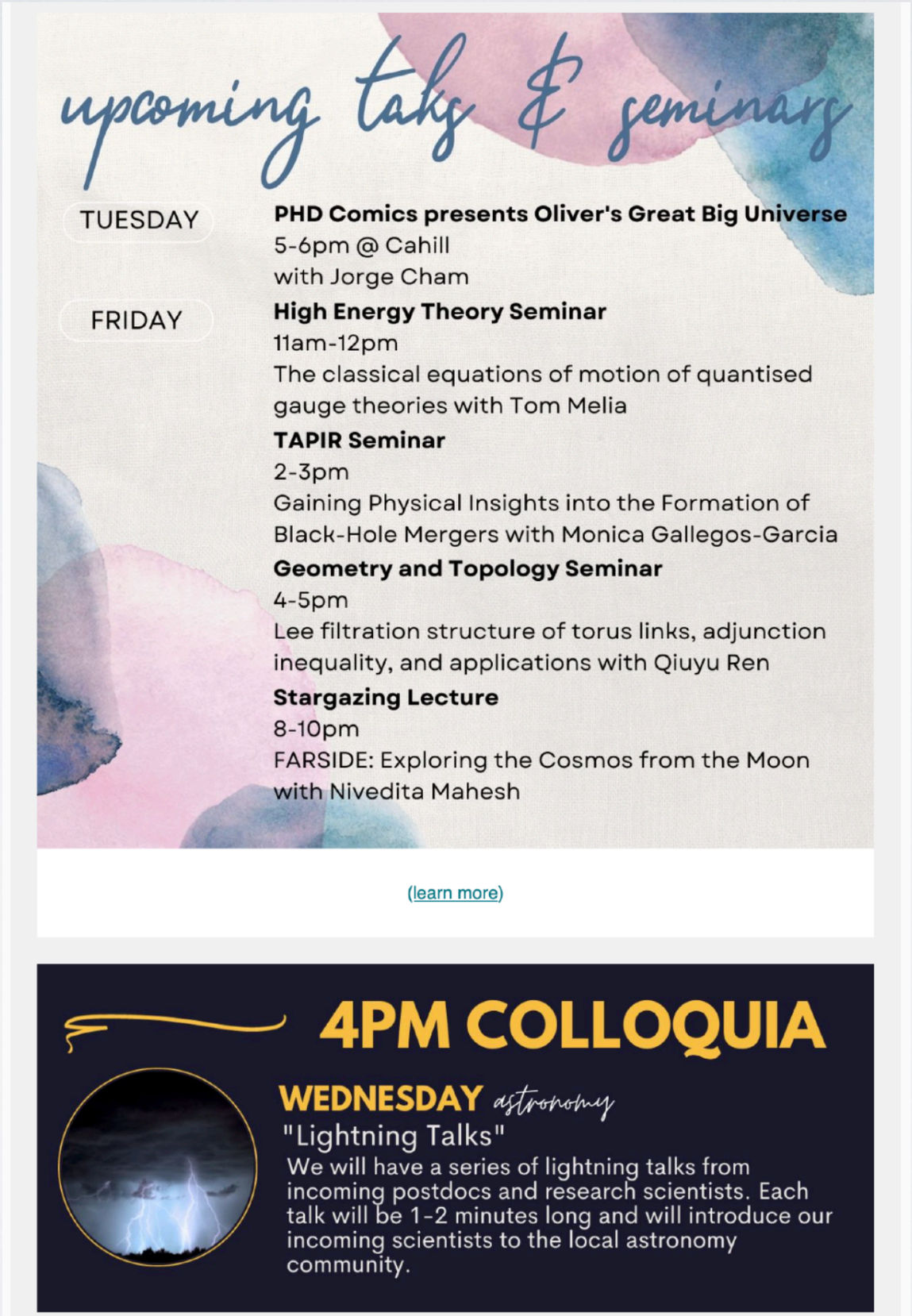
Sponsors

We are pleased to partner with Caltech Theater and Public Programming for the event. We also thank the following generous sponsors for making the competition possible.

Diamond Level

EBSCO Information Services

1-Minute Lightening Talk @ Caltech



15-Second Elevator Pitch

